

Session 6.1

From Chat Window to Agentic AI: What an Agent Is, and Where It Pays

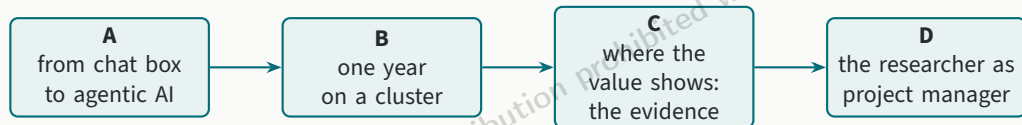
Part one of five: chatbot, copilot, agent, agent team — what makes AI agentic, and one year of an agent on a cluster, in the instructor's own projects



Zhigang Feng

The Question, and the Route

What is an agent, what makes AI agentic —
and what did it change for one researcher?



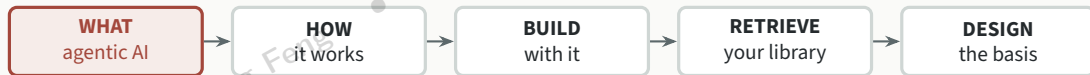
- ▶ Session 5 ended with a language model you can score. Everything it produced was **advice**: you still ran the code, moved the files and checked the numbers.
- ▶ This session closes the gap between advice and action — and opens with a year in which it closed on one account.
- ▶ **Four parts: what an agent is, what it did on a cluster, what 637 scientists report, and what the job becomes.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §1; Codreanu et al. (2026).

Session 6.1

From Chat Box to Agentic AI

What makes AI agentic, and what does not change



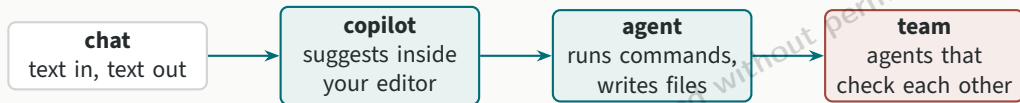
From a Chat Box to Agent Teams, 2022–2026: The Rows Deck 1.3 Left Out



- ▶ Since Deck 1.3 drew this: models that think before they answer (OpenAI o1-preview, Sep 2024); the protocol moved to a Linux Foundation body (Dec 2025); task length doubles about every 89 days (METR, Jan 2026).
- ▶ **Each step made the work cheaper; none made checking it cheaper.**

Source: Deck 1.3's figure; rows added from Anthropic Economic Index (28 Apr 2025), Linux Foundation (9 Dec 2025), METR Time Horizon 1.1 (29 Jan 2026), Lu et al. (2026, Nature), Codreanu et al. (2026), all read 23 Sep 2026; OpenAI (12 Sep 2024).

Chat, Copilot, Agent: A Spectrum, Not a Switch



what changes across the strip is not the model — it is how much of your machine it can touch

- ▶ One measured point on the strip: in April 2025, 79% of conversations with a terminal agent were *automation* (the model does the task) against 49% in the chat window — the same index Deck 1.2 used to place AI inside occupations.
- ▶ A chat window is still the right tool for one question and one answer.
- ▶ **You choose a position on this strip for each task, and you can move left when a task turns out to be delicate.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §3.1–§3.2; Anthropic Economic Index, “AI’s impact on software development”, 28 Apr 2025 (read 23 Sep 2026).

What Makes AI Agentic: Four Marks, All Needed

tool use

calls Python, R,
Stata, the shell

autonomy

a goal, not
a prompt: it
plans the steps

iteration

reads the error,
fixes, reruns

persistence

conventions and
plans kept on disk

An **agent** is a model in a loop, given tools and pointed at a goal: the cleverness is in the model, the *capability* in the loop.

- ▶ The marks are cumulative. Tool use without autonomy is a console; autonomy without iteration dies at the first error; without persistence nothing survives the session.
- ▶ **“Agentic” is a degree, not a type: how much of the loop runs without you.**

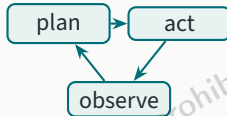
Source: Miao & Feng, *AI for Economists*, ch. 15, §3.1.

Agent, Workflow, Agent Team: Who Decides the Next Step

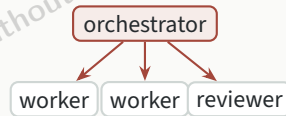
a workflow: code decides



an agent: the model decides



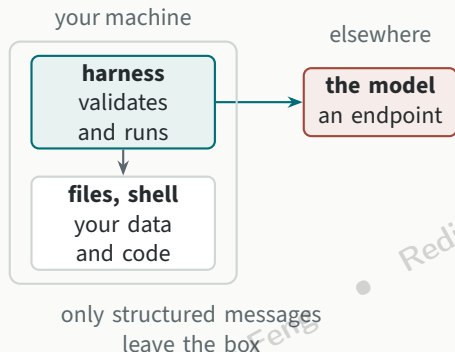
a team: agents split the work



- ▶ In a workflow, models and tools are “orchestrated through predefined code paths”; in an agent, models “dynamically direct their own processes and tool usage”.
- ▶ A workflow is cheap and predictable; an agent handles surprises; a team buys breadth and pays in coordination.
- ▶ The advice is to start at the left: “finding the simplest solution possible, and only increasing complexity when needed”.
- ▶ **Capability and risk both sit with whoever chooses the next step.**

Source: Anthropic, “Building effective agents”, 19 Dec 2024 (read 24 Sep 2026); Miao & Feng, *AI for Economists*, ch. 15, §3.1 and §4.3.

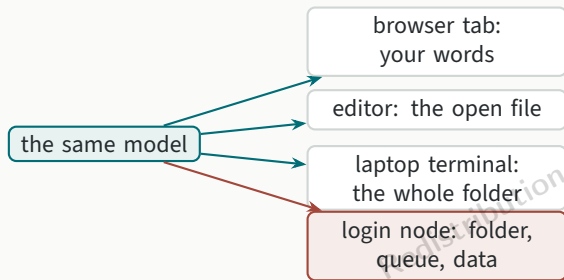
Three Parts, and Only One of Them Is Clever



- ▶ **The harness:** an ordinary program on your computer. It reads and writes files, runs commands, and enforces the rules you set.
- ▶ **The model:** not on your computer. An endpoint that reads messages and names the next call. That is the whole of it.
- ▶ **Files and tools:** your data, your code, your shell — all local.
- ▶ **The intelligence sits in one part; the discipline sits in the other two.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §3.2.

Same Model, Different Reach: Where the Loop Runs

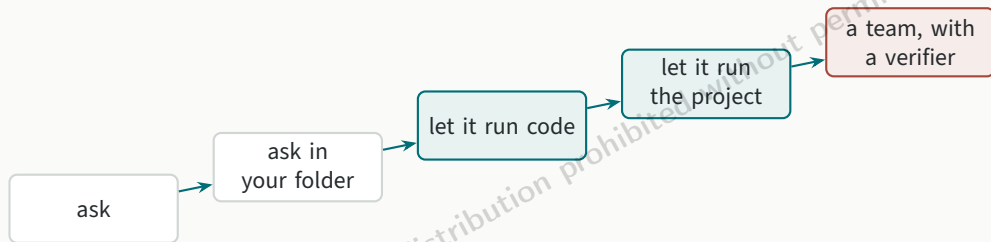


The capability that matters for research is not the model's reasoning. It is the *blast radius* of what it is allowed to reach — and the fourth box is Part B's whole story.

- ▶ The reach dial is where the harness runs, not which model answers: on a login node it reaches the folder, the scheduler, and data that never has to move.
- ▶ **“Install the AI on the server” was not a new capability. It was the same loop, moved to where the files are.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §3.2; this deck's Part B, by role.

The Ladder: From a Chat Window to a Verified Team



climb one rung only when
the rung below is boring

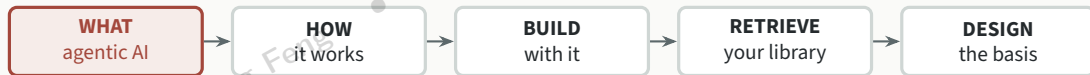
- ▶ **Every rung up buys throughput and costs verification. The last rung is the only one where an agent checks an agent.**
- ▶ The year on the cluster reached the fourth rung; Deck 6.3 builds the fifth.

Source: Miao & Feng, *AI for Economists*, ch. 15, §4.3.

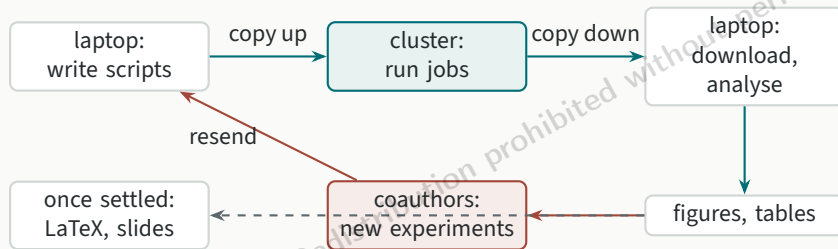
Session 6.1

One Year on a Cluster

The loop before, the loop after, and what the agent does now



The Pre-AI Loop: Write Locally, Run Remotely, Analyse Locally

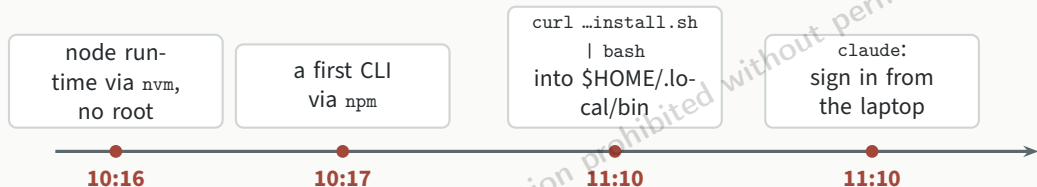


every arrow was a manual transfer; analysis and writing never touched the cluster; the loop closed weekly at best

- **Two machines, one person carrying files between them. The cluster computed; everything else happened where the researcher sat.**

Source: Instructor's project record (the national-debt pipeline), 2025–26, by role.

Install Day: Three Commands on a Login Node, 20 Apr 2026

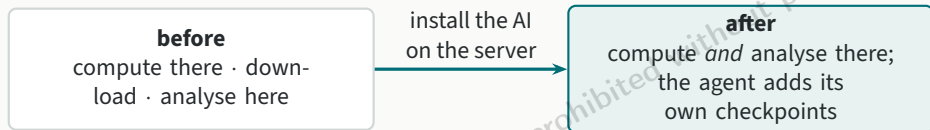


the harness's own state file records the first start at 11:10:22 that morning; by 23 Sep 2026 it counted 224 startups across 24 project folders

- ▶ No administrator was involved: a user-local runtime, a user-local binary, a browser sign-in on the laptop.
- ▶ **The whole install was under an hour on a login node, and it is the least interesting fact in this deck.**

Source: Instructor's shell log and harness state file, read 23 Sep 2026; times in the cluster's zone.

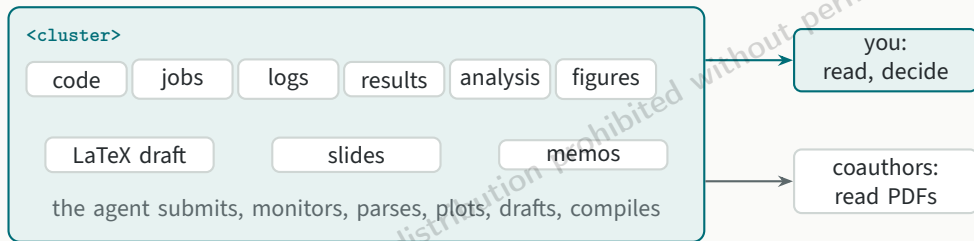
The Aha, on Tape: 5 May 2026



- ▶ “Before, I had it run the computation on the server, then I downloaded the results and analysed them myself.”
- ▶ “Then, almost by accident last week, I installed the AI on the server itself and let it operate there, adding its own checkpoints — and a lot of problems went away.”
- ▶ **The change was not a better model. It was where the loop ran.**

Source: The instructor, on a coauthor call, 5 May 2026; the instructor’s own words, translated. Nothing of the coauthor’s is shown.

The New Loop: Everything Happens Where the Data Is



- ▶ One machine, one folder tree, one agent that runs from the project's root. Nothing is copied down to be analysed; what leaves the cluster is a PDF.
- ▶ **The loop that closed weekly closes the same day, and the researcher's part of it shrank to two verbs.**

Source: Instructor's project record, 2026, by role. A course-recorded run on the same cluster is left for the instructor to record; no number from it is printed here.

One Project, End to End: The National-Debt Pipeline



each arrow is a command the agent runs from the project root;
the draft's numbers trace to job outputs; the coauthor reads PDFs

- ▶ The model is a model of government debt and default, solved by sweeps of C++/MPI jobs; the 2026 revision programme was 39 jobs, designed from a coauthor's comments and submitted within days of them.
- ▶ The project's standing brief lists every script and what it does, so the agent can rerun the chain without being told how.
- ▶ **The unit of instruction became a project state (“follow the plan; run the new experiments there”), not a question.**

Source: Instructor's project repository, read 23 Sep 2026; paths symbolic; no result shown.

Four Real Prompts, Redacted

4 May 2026, a literature project [monitor]

> i just worried that <cluster> may kill the jobs if they run more than 30 minutes or so. keep an eye on that. keep doing what you are doing.

25 Jul 2026, the national-debt project [execute the plan]

> Then follow the plan, make changes. For new experiment, just keep in mind, I will run them on a different cluster, <cluster>. See here for previous script how we run experiments. <submit script>

25 Jul 2026, the national-debt project [diagnose]

> mkdir: cannot create directory '<spool path>': Permission denied.
all jobs failed due to errors above.

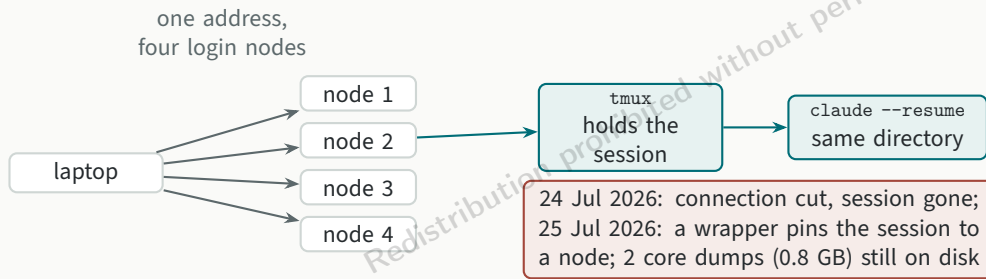
25 Jul 2026 [govern the tool]

> use job-specific scancel <jobid> instead of scancel -u <user>, so stops killing unrelated jobs.

- **The verbs are monitor, execute, diagnose, govern. None is a question; each hands over a state of the project and a rule.**

Source: Instructor's harness prompt history, 2026; typed as shown; names of hosts, scripts and paths redacted.

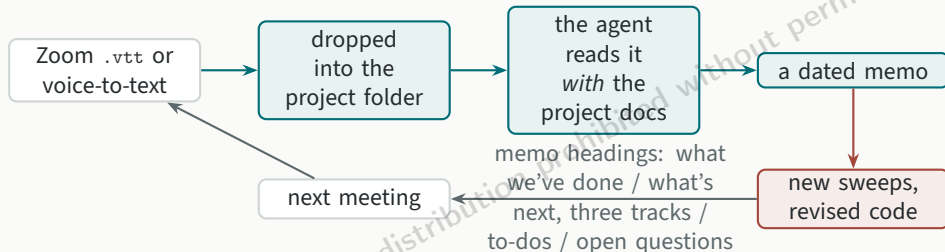
Things Break: Surviving a Disconnect



- ▶ “The problem is when the connection to cluster was cut off ... I cannot resume the previous session. Is there any way to avoid that?”
- ▶ **A session is a list on disk (Deck 6.2), so it resumes from the same directory — but only on a node you can find again.**

Source: Instructor’s notes file and wrapper script, 24–25 Jul 2026; harness history; counts by role, 23 Sep 2026.

The Meeting Loop



one instance, by file times: transcript 02:29 → memo 03:44 → document 03:46

- ▶ The memo is a file in the repository, so the next experiment is designed from a document, not from memory.
- ▶ **Seventy-five minutes from a recording to a memo the coauthor can read — and the memo names the next sweep.**

Source: Instructor's memos of 5 May and 7 Jun 2026 and one interview pipeline of 20 Jun 2026; headings and file times only.

From Results to a Draft in One Day

status memo

- 1 TL;DR
- 2 cleanup
- 3 toolchain
- 4 sweep inspection
- 5 rerun (job ids)
- 6 next steps

a 57-page
computational draft,
10 Sep 2026

its provenance rule:
“no number appears in
the paper that is not
traceable to a frozen artifact”

the memo's results table is shown with its cells blanked:
what matters is that a table with job ids and pre-flight checks exists before a sentence of the draft does

- ▶ The draft was written from a plan file in a day; every figure regenerates from frozen artifacts by a script.
- ▶ **Speed is not the point: a fast draft is only safe when every number has an address.**

Source: A structural-model project's records, by role: memo structure, page count, date and the rule; no result shown.

The Inventory: What the Agent Does for Me Now

coding	mathematical analysis	numerical design	empirical analysis
editing drafts	formatting slides and drafts	publishing the website	literature review
IT: job errors, package installs	meeting memos	continuous learning: the tool teaches its own failure modes	

- ▶ IT without a ticket: a submission error parsed and fixed, a CPU job that must still request a GPU, a TeX distribution installed into project space — each one prompt and one afternoon.
- ▶ **The mundane list is the biggest saving; the last item is the one that changes the researcher.**

Source: Instructor's own account, Aug 2026; a status memo of 9 Jun 2026, \$3, and a job script; artifacts by role.

Where the Time Went: One Week, Before and After

before

write and debug code

move files, wait

analyse

write

meet

after

specify

read diffs, logs, drafts

decide, verify

write

meet

scale on the account, recounted 23 Sep 2026: 324,394 lines of Python, 135 experiment folders, 6,229 checkpoint files across two computational projects

- ▶ The strips are the instructor's description, not a time-use survey; the counts are the recount, with their definitions.
- ▶ **What grew is not the number of experiments alone. It is the share of the week spent reading what a machine produced.**

Source: Instructor's Aug 2026 statement, recounted on 23 Sep 2026 by role.

Continuous Learning: The Log as a Textbook

a project log, 861 KB

139 entries, dated

...

2026-07-22 — probe verdict

2026-07-22 — probe 2

FAILED fast

2026-07-24 — gate FAILED,
cause identified

SUPERSEDED — read the
resume file first

every correction
becomes a file

the tool learns the
project's conventions

the researcher learns
the tool's fail-
ure modes

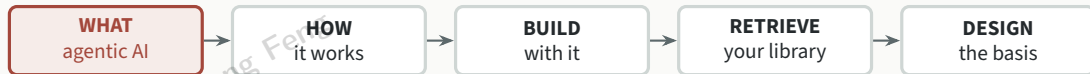
- ▶ Entries in capitals are verdicts: a probe that failed, a gate that failed, a plan that was superseded. They are described here, not quoted.
- ▶ **The log is where the two kinds of learning meet, and it is the artifact a new coauthor reads first.**

Source: A project log, by role; counts on 23 Sep 2026.

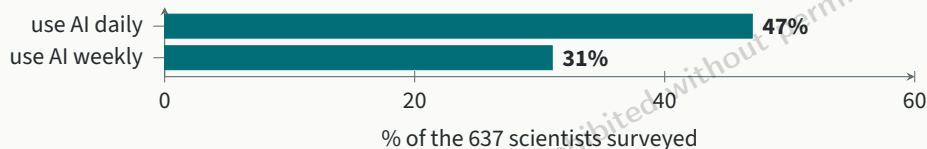
Session 6.1

Where the Value Shows

637 scientists said where their saved time went — Session 1's survey, re-read with one question



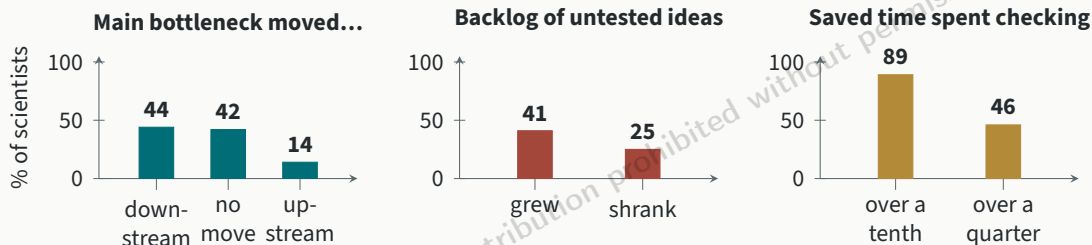
What a Sample of 637 Scientists Reported



- ▶ Nine days ago, in Deck 1.1, this survey showed that scientists use AI far more than other occupations. Today it is re-read with one question: *where did the saved time go?*
- ▶ The base: 637 active researchers in the US and UK, fieldwork 27 July to 11 August 2026; a screened non-probability sample.
- ▶ 47% use some form of AI daily and another 31% weekly. Adoption is no longer the question.
- ▶ **The next two frames are Session 1's, redrawn as they were; each carries one new bullet, and it is this deck's.**

Source: Codreanu, Imas, Mateos-Garcia et al. (2026), *AI in Science: Early Insights*, pp. 4, 23, 42; read 23 Sep 2026.

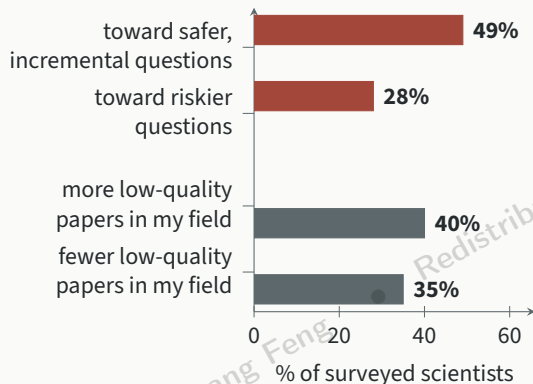
Findings 3–4: Time Saved, and Where the Bottleneck Went



- ▶ Reported saving: 6.9 hours a week; just under three quarters save time on net, 6% lose it. About 70% of it returns to research: more output (just under 30%), lab work (21%), harder problems (19%).
- ▶ **The verification tax: of those who save time, 89% spend over a tenth of the saving checking AI output, and 46% over a quarter.**
- ▶ **Part D's first danger, in survey numbers: the queue moved to the checker's desk.**

Source: Codreanu et al. (2026), pp. 5, 23–26 (their Figure 13). Third panel: those who report saving time.

A Warning Inside the Same Data



- ▶ 84% report more output over three years. Output of what kind?
- ▶ The authors' reading: perhaps a "Streetlight Effect". AI cheapens work where data and benchmarks already exist.
- ▶ Top bottleneck today: physical experiments and data collection (24%). No language model runs a laboratory.
- ▶ **More output is not the same as more discovery.**

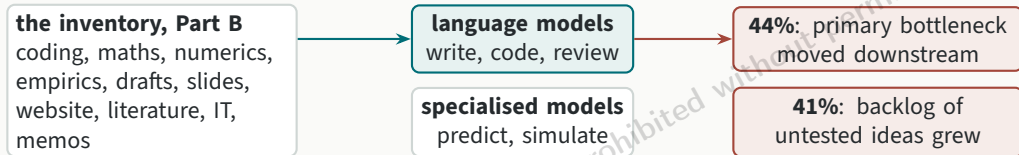
Source: Codreanu et al. (2026), pp. 24, 26, 30.

How to Read This Evidence

- ▶ The usage data are the vendor's own telemetry. Enterprise accounts and agentic tools are not observed.
- ▶ The survey is a screened non-probability sample: unweighted, with no margin of error quoted.
- ▶ Enthusiasts answer surveys about AI. The authors say this may overstate adoption and time saved.
- ▶ Every quantity is self-reported, and the results “establish associations rather than causality”.
- ▶ **What the data do support:** scientists use AI far more than other occupations, and they report a constraint moving from generating ideas to testing and checking them.
- ▶ **Self-reports against a measurement** (METR, Jul 2025): developers forecast 24% faster, believed 20%, measured 19% *slower*, 9% of time reviewing AI output; METR's Feb 2026 follow-up (−4%, −15 to +9) is, it says, an unreliable signal — selection.

Source: Codreanu et al. (2026), §6.1 (limitations), pp. 28–29, and appendix, p. 42.

The Cluster Story in Their Numbers



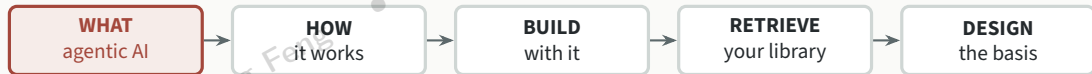
- ▶ The report's two families: language models that write, code and review; specialised models that predict and simulate. Every item on Part B's inventory is in the first; the solver is C++, not a model.
- ▶ 44% say their primary bottleneck moved downstream in two years; 41% say their backlog of untested hypotheses grew. Part B's meeting loop ended in "new sweeps, revised code".
- ▶ **"Suggest new sweeps" is the backlog: proposals arrive faster than one researcher can judge them. One account, one draw from this population — and not an outlier.**

Source: Codreanu et al. (2026), pp. 4–5, 25; Part B's inventory and meeting-loop frames.

Session 6.1

The Researcher as Project Manager

What the job became, and two dangers named now



The Bottleneck Moves; Four Judgements Do Not

is the *question*
worth asking?

is the *model*
the right one?

is the *design*
credible?

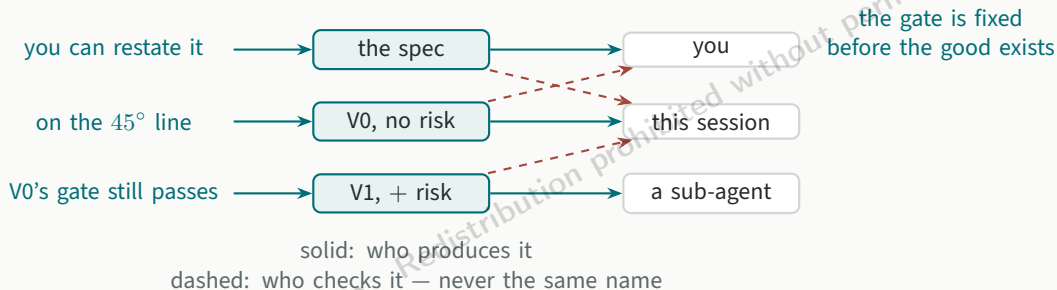
is the *result*
believable?

None of the four is cheaper than it was. Each is now a larger share of what you are paid for.

- ▶ Deck 1.1 said it in prices: when one input becomes elastic, its complements bind. The four judgements are the complements.
- ▶ Execution was the bottleneck: the weeks between a good idea and a first number. When that cost falls, the scarce input is knowing which numbers to believe.
- ▶ **In the year on the cluster, the researcher's two verbs became “read” and “decide” — and both got longer, not shorter.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §4.1; Deck 1.1's frame on elastic inputs.

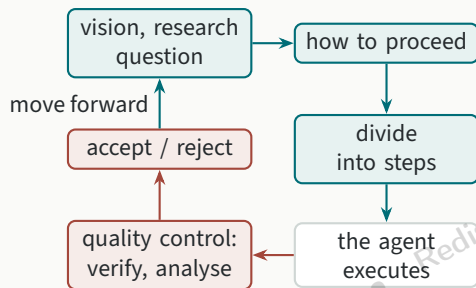
You Are the Project Manager



- Break the goal into **intermediate goods** — a specification, a validated V0, a cleaned panel, a review report — and fix each good's gate *before* it is produced.
- **Delegating is a principal-agent problem, so every good gets a gate, an owner, and a reviewer who did not build it.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §4.1 and §4.4, the agent workflow.

The Job Description



A **gate** is a bound fixed before the work exists: a number to reproduce, a test to pass, a property to hold. Written after the fact it is not a gate but a rationalisation.

- ▶ Six of the seven boxes were always the researcher's. What changed is that the seventh no longer takes the weeks.
- ▶ **The job is the cycle, and the cycle is management.**

Source: Instructor's account, 2026; Miao & Feng, *AI for Economists*, ch. 15, §4.1.

A Paradigm Shift: What Changed, What Did Not

What changed (the national-debt project's record)

- ▶ The comment–revise–rerun loop compressed: a coauthor's comments of 22 Jul, a response plan by 24 Jul, 39 jobs submitted from 25 Jul.
- ▶ Documentation became executable: the standing brief lists every script and what it does, and the agent runs from it.
- ▶ The unit of progress became the *validated milestone*: a sweep that passed its checks, not a week that passed.

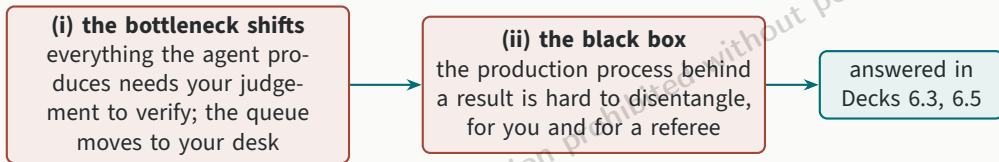
What did not change

- ▶ The AI never decided what was economically interesting.
- ▶ It never caught the wrong timing on its own.
- ▶ It never knew when a result was not credible.

The machine moved every bound except the one that defines the researcher.

Source: Instructor's project record, file dates read 23 Sep 2026; the right column from the instructor's earlier lecture notes.

Two Dangers, Named Now and Answered at the End

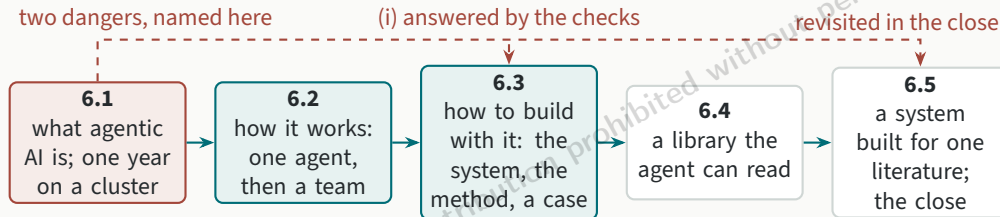


the survey evidence for (i) is Part C: 637 scientists, and where their saved time went

- ▶ Both are real on the account described here; neither is an argument for going back.
- ▶ **For now the priority is to understand how to leverage the rapid advance so that it works for us — and to name what it costs.**

Source: Instructor's account, Sept 2026; Codreanu et al. (2026) as the pointer to the evidence.

The Session Map: Five Decks, One Arc



- ▶ Decks 6.1 to 6.3 follow one thread: what makes AI agentic, how it works, how to build with it. Then a library the agent can read (6.4), and a system built for one literature, with the close (6.5).
- ▶ **Deck 6.2 next: a loop you could write yourself, and the four marks of an agent as lines of code.**

Source: The session's five decks; Miao & Feng, *AI for Economists*, ch. 15.